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इलेक्ट्रॉनिकी एवं सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF ELECTRONICS AND INFORMATION TECHNOLOGY
सत्यमेव जयते

Chips to Startup Programme

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)
BAJAJ INSTITUTE OF TECHNOLOGY, WARDHA

TEAM NAME : SYNTAX SQUAD

PROBELM STATEMENT NO : PS09

PROBLEM STATEMENT TITLE : INT8 Quantized CNN Deployment for Resource-Constrained Microcontrollers

PS Category- Software

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PROBLEM STATEMENT

- **Deep learning models are too large to run on tiny, low-power microcontrollers used in IoT and embedded devices.**
- **On-device AI removes cloud dependency, cuts latency, and protects privacy — if the model fits in a few hundred KB.**
- **Typical MCUs offer only 256KB-1MB flash and 64256KB RAM, while trained FP32 models often run into several MB.**

YOUR SOLUTION

- **Project: Edge Quant CNN**
- **A CNN training + INT8 quantization pipeline that shrinks trained models ~3-4x with near zero accuracy loss.**
- **Trains CNN >> converts to TF Lite >> applies INT8 Post-Training Quantization >> verifies FP32 vs INT8 accuracy/size >> deploys and runs inference inside a QEMU-emulated microcontroller using TensorFlow Lite Micro.**

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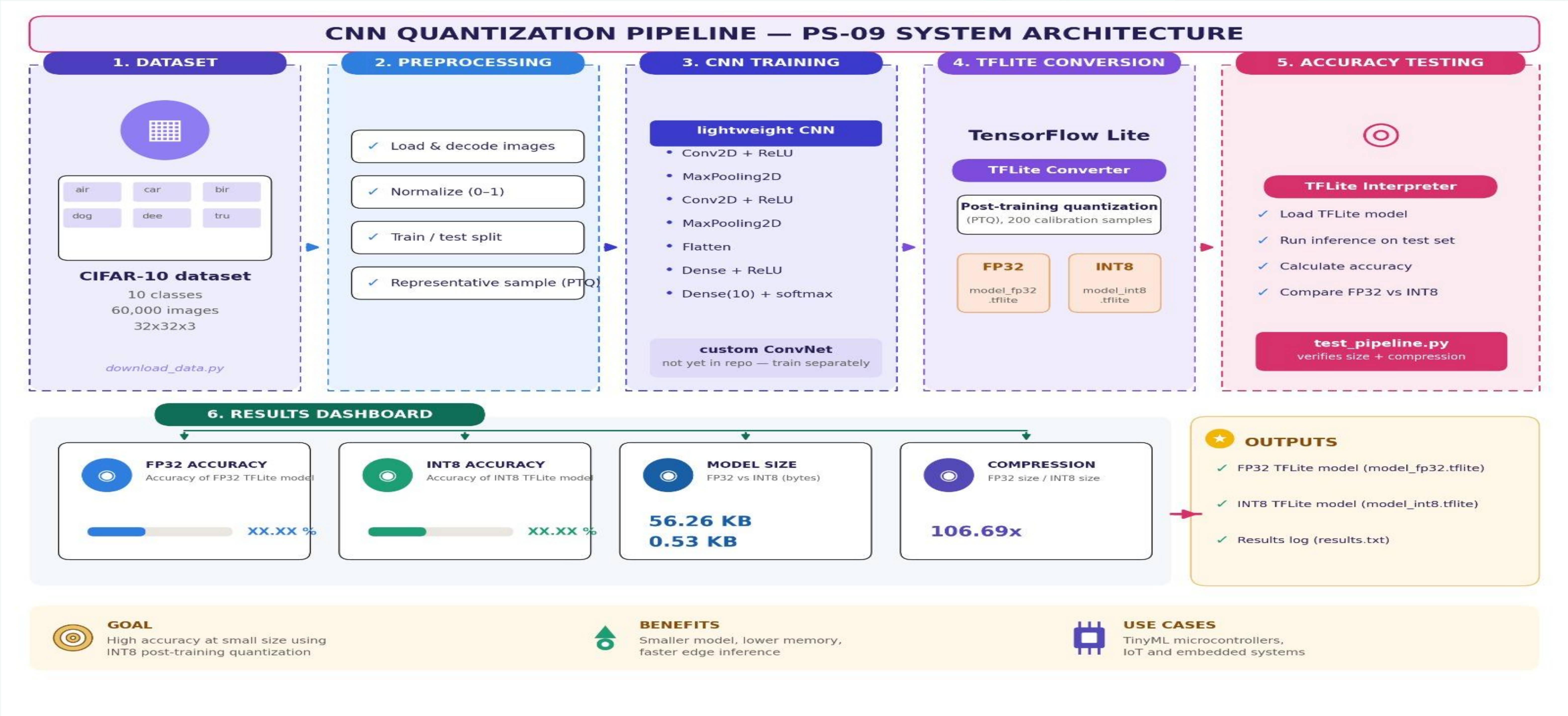
ARCHITECTURAL BLOCK DIAGRAM



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PURPOSE OF BLOCKS

1. INPUT BLOCK — Dataset & Preprocessing

Idea: CIFAR-10 (10 classes, 60K images) — standard, small dataset, fast training. Steps: normalize, train/test split, representative sample for PTQ calibration.
Trade-off: Low resolution (32x32) → fast training, but real-world accuracy may be lower.

2. PROCESSING BLOCK — CNN Training + TF Lite Conversion

Idea: Lightweight CNN chosen for edge deployment — small size, fast inference. Post-Training Quantization (PTQ, 200 samples) used since it needs no retraining, simpler to implement.
Trade-off: Lightweight model → lower accuracy ceiling. PTQ is simpler/faster than QAT, but causes bigger accuracy drop after INT8 quantization. Only 200 samples may not fully represent data → risk of calibration error.

3. OUTPUT BLOCK — Accuracy Testing + Results Dashboard

Idea: FP32 vs INT8 comparison gives direct proof of quantization effect (accuracy, size, compression). Using TF Lite Interpreter for real inference makes testing reliable.
Trade-off: 106x compression is impressive, but actual accuracy drop must be clearly measured (currently placeholder). Extreme size reduction (0.53 KB) can risk meaningful accuracy loss — needs verification for the use case.
Overall: Pipeline balances **accuracy vs efficiency** — optimized for small size and fast edge inference, at the cost of some accuracy.

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OUTPUTS AND RESULTS

```
(veltraxx_env) PS C:\Users\LATHIKA CHIDAMBARAM\Desktop\hackthon> python test_pipeline.py

==========================================
         FP32 vs INT8 RESULTS
==========================================
FP32 Model Size : 56.26 KB
INT8 Model Size : 0.53 KB
Compression     : 106.69x
==========================================
```

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RESEARCH AND REFERENCES

- **TensorFlow Lite Post-Training Quantization –** **https://www.tensorflow.org/lite/performance/post_training_quantization**
- **TensorFlow Lite for Microcontrollers (TFLite Micro) –** **https://www.tensorflow.org/lite/microcontrollers**
- **CIFAR-10 Dataset (Krizhevsky, 2009) – https://www.cs.toronto.edu/~kriz/cifar.html**

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